

Supplementary Material

Effects of Mn Promotion on the Activity and Selectivity of Co/SiO₂ for Fischer-Tropsch Synthesis

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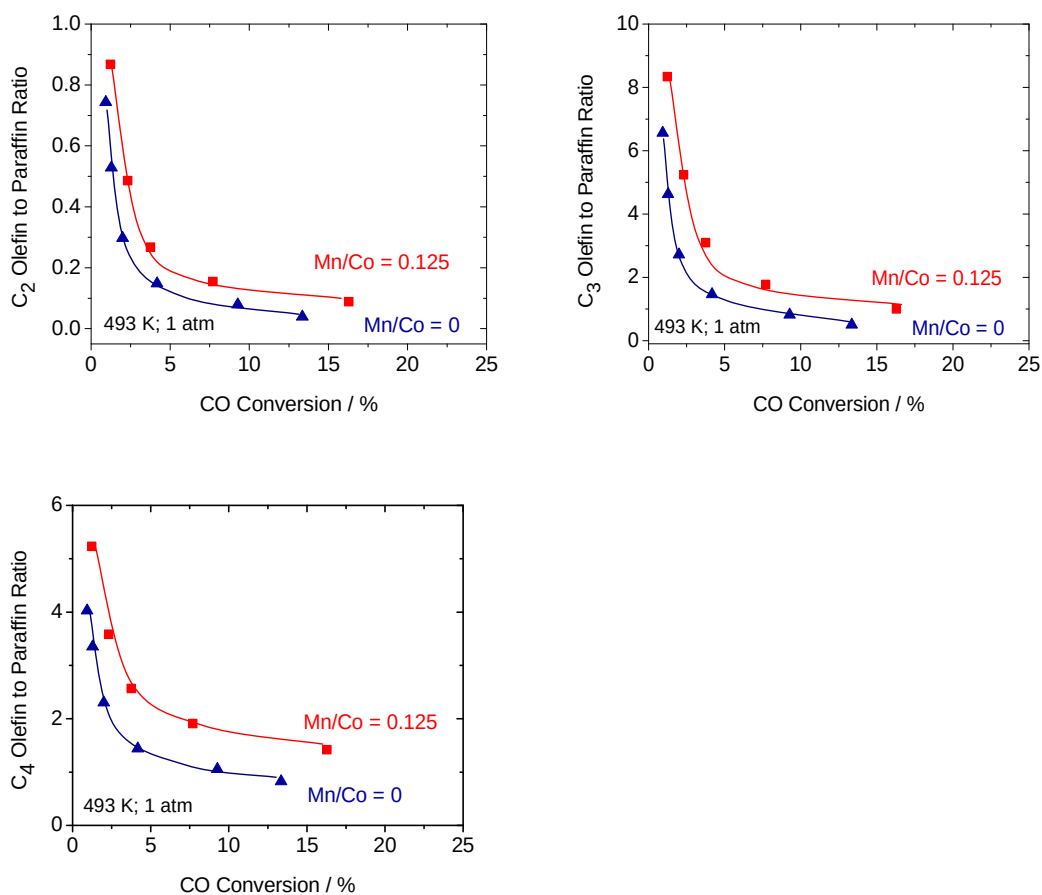


Fig. S1: Effect of CO conversion on C₂ (left), C₃ (right) and C₄ (bottom) olefin to paraffin ratios at 493 K and 1 atm for catalyst with Mn/Co ratios of 0 and 0.125. Feed ratio H₂/CO/Ar = 2/1/0.2, flows = 2.5 to 100 ml/min, catalyst mass = 100 mg.

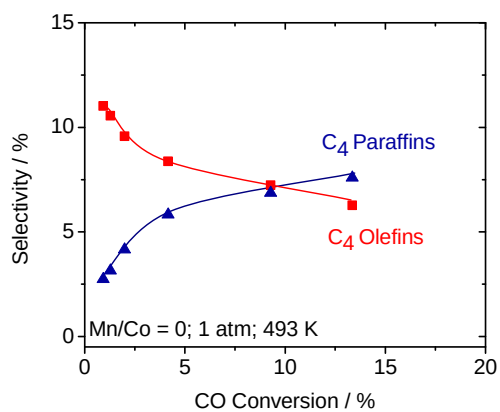
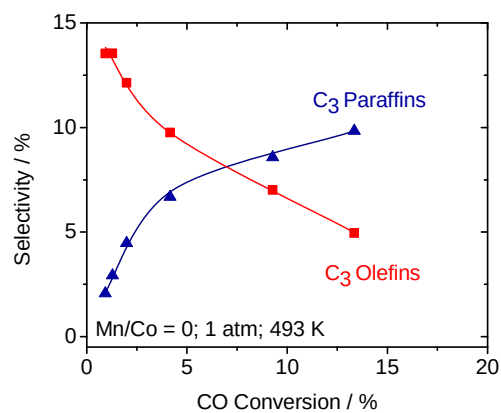
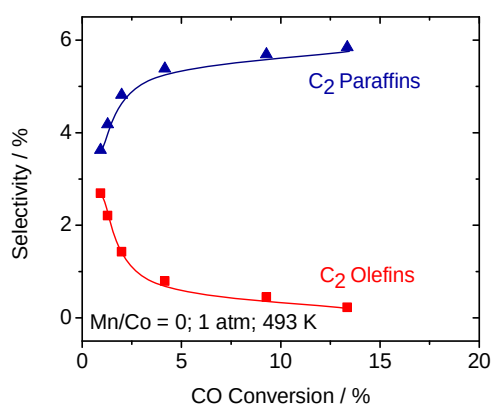


Fig. S2: C₂ (left) C₃ (right) and C₄ (bottom) olefin and paraffin selectivity as a function of increasing CO conversion at 473 K and 1 atm for catalysts without Mn promotion. Feed ratio H₂/CO/Ar = 2/1/0.2, flows = 2.5 to 100 ml/min, catalyst mass = 100 mg.

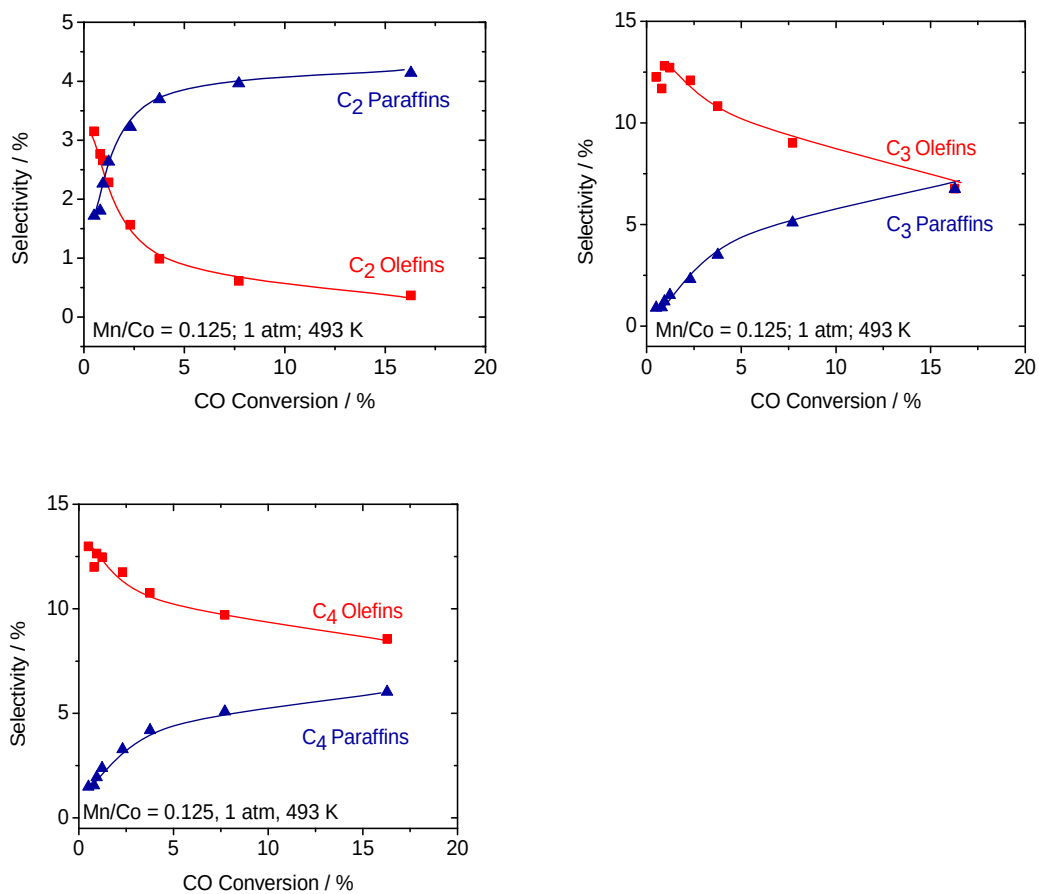


Fig. S3: C₂ (left) C₃ (right) and C₄ (bottom) olefin and paraffin selectivity as a function of increasing CO conversion at 473 K and 1 atm for Co/SiO₂ with a Mn/Co ratio of 0.125 (right). Feed ratio H₂/CO/Ar = 2/1/0.2, flows = 2.5 to 100 ml/min, catalyst mass = 100 mg.

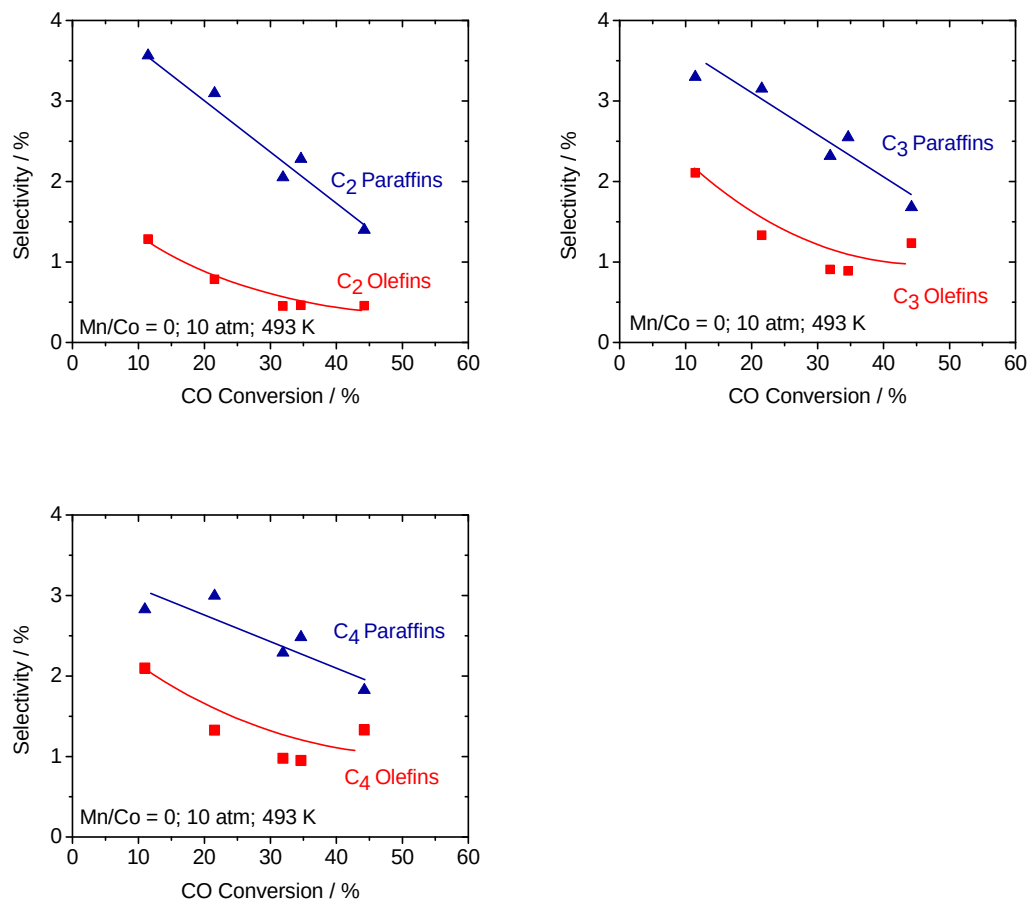


Fig. S4: Effect of pressure on olefin and paraffin selectivities within the C₂ C₃ and C₄ fraction at varying degrees of CO conversion for unpromoted Co/SiO₂. Temperature = 493 K, feed ratio H₂/CO/Ar = 2/1/0.2, flows = 20 to 120 ml/min, catalyst mass = 100 mg.

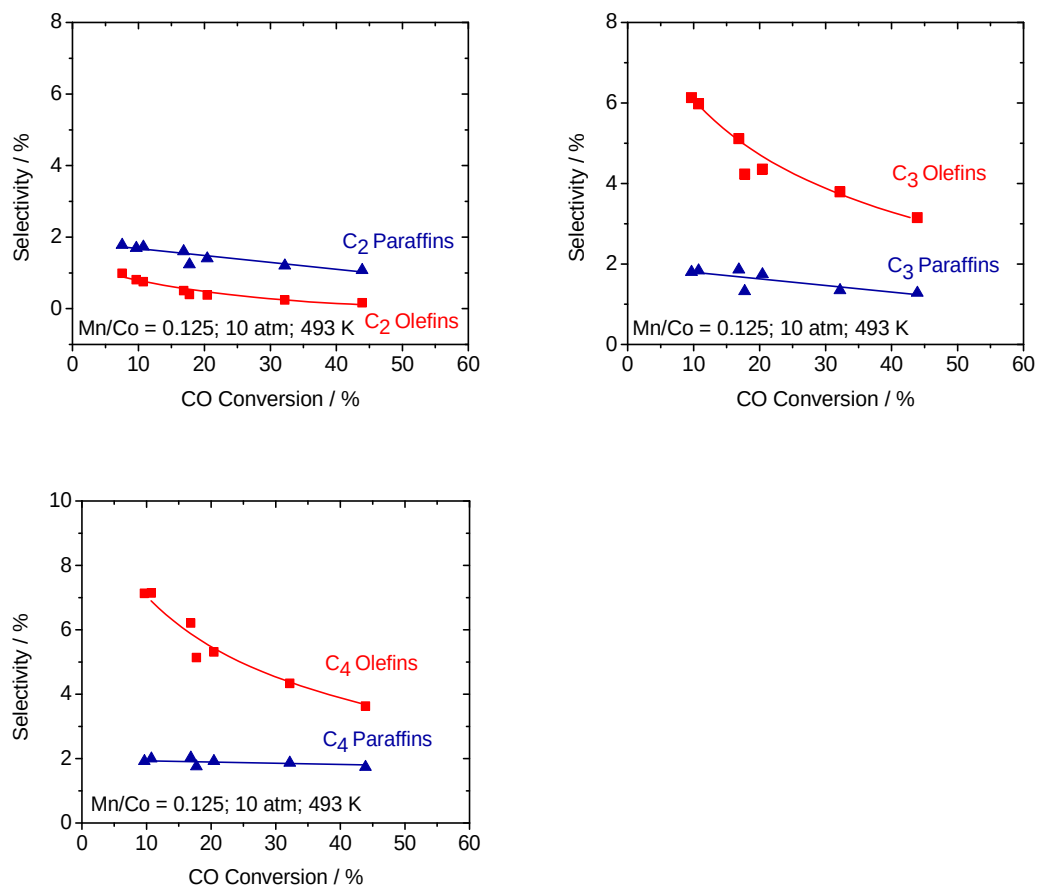


Fig. S5: Effect of pressure on olefin and paraffin selectivities within the C₂ C₃ and C₄ fraction at varying degrees of CO conversion for Mn promoted Co/SiO₂ with Mn/Co ratios of 0.125. Temperature = 493 K, feed ratio H₂/CO/Ar = 2/1/0.2, flows = 20 to 120 ml/min, catalyst mass = 100 mg.

Mn/Co	Conversion %	Selectivity / %				Ratio α /non- α
		1-butene	cis-2-butene	trans-2-butene	butane	
0	5.7	3.2	2.7	2.1	5.3	0.7
0	25	2.0	1.9	1.3	7.8	0.6
0	39.1	1.9	1.6	1.0	8.3	0.7
0.125	6.6	5.7	2.2	2.0	3.8	1.3
0.125	25.5	3.0	3.0	2.2	6.1	0.6

Table S1: Effect of CO conversion on the formation of α and internal olefins within the C₄ fraction for unpromoted and Mn promoted Co/SiO₂ with Mn/Co ratios of 0.125 at 1 atm. Temperature = 493 K, feed ratio H₂/CO/Ar = 2/1/0.2, flows = 20 to 120 ml/min, catalyst mass = 100 mg.

Mn/Co	Conversion %	Selectivity / %				Ratio α /non- α
		1-butene	cis-2-butene	trans-2-butene	butane	
0	5.9	0.9	1.1	0.7	1.5	0.5
0	24.8	0.5	1.9	0.6	0.4	0.2
0	46.0	0.4	0.5	0.4	1.8	0.4
0.125	7.8	5.7	0.3	0.2	1.6	10.9
0.125	20.4	4.6	0.3	0.4	1.9	7.0
0.125	43.9	2.9	0.3	0.4	1.7	4.0

Table S2: Effect of CO conversion on the formation of α and internal olefins within the C₄ fraction for unpromoted and Mn promoted Co/SiO₂ with Mn/Co ratios of 0.125 at 10 atm. Temperature = 493 K, feed ratio H₂/CO/Ar = 2/1/0.2, flows = 20 to 120 ml/min, catalyst mass = 100 mg